

FINLEY HARRISON

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EDUCATION

University of Waterloo

2023 - Present

Bachelor's of Computer Science Honours Co-op Program

GPA: 3.8

Courses: Data Structures and Algorithms, Intro to ML, Operating Systems, Database Systems, Object-Oriented Programming, Computer Networks, Probability & Statistics, Linear Algebra, Computer Architecture, Compilers

TECHNICAL SKILLS

Languages	C/C++, C#, Java, Python, JavaScript, TypeScript, SQL, Bash, Racket
Frontend	React, Next.js, Three.js, Tailwind CSS, Redux, jQuery, HTML/CSS
Backend/Cloud	ASP.NET, Node.js, Flask, Azure (EventHub, ADX), Supabase, Appwrite, PostgreSQL
Tools/DevOps	Git, Docker, Linux, Ollama, MCP, RAG, Nginx, Jira, Confluence, Blender, Adobe Suite

EXPERIENCE

Rocscience Inc.

Jan 2026 – Apr 2026

Software Engineer Intern

Toronto, ON

- Engineered an Audit Dashboard using **ASP.NET Core/SQL** to monitor system-critical APB log transactions.
- Deployed a scheduled batch cleanup process for stale tenant data, reducing database size by **30%**.
- Optimized performance by **refactoring legacy C#** methods, achieving up to **5x speedups** in critical code paths.
- Maintained high-velocity output, resolving **90+ tickets** spanning backend bug fixes and UI enhancements.
- Automated report validation via **CI/CD scripts and 500+ unit tests**, ensuring integrity for engineering datasets.

BCS Automation Ltd.

May 2025 - Aug 2025

Software Engineer Intern

Belleville, ON

- Had a lead role in designing and implementing an analytics dashboard using **React** and **C#**, **ASP .NET**, enabling live monitoring of 5+ Coast Guard ships and processing over **50K+** telemetry points minimizing data loss.
- Architected a real-time data pipeline using Azure EventHub and ADX to ingest **2GB+/hr** of telemetry data.
- Designed and trained ML models for **real-time anomaly detection** and **forecasting**, achieving over **90%** accuracy.
- Streamlined internal documents with a chatbot (Ollama, MCP, RAG), reducing information retrieval time by **60%**.
- Implemented Redis caching to frequently used endpoints, cutting API response times from **3.2s to 120ms**.

University of Waterloo IST

May 2024 - Aug 2024

Service Desk Specialist

Waterloo, ON

- Spearheaded the development of an AI chatbot to resolve user issues, driving a **20% decrease** in support requests.
- Engineered an automated script to clear and configure new Windows accounts, saving the team **3+ hours weekly**.
- Effectively managed **400+ weekly** technical support requests from students and staff via phone and Jira.
- Leveraged communication skills to secure the term's **highest client reviews**, consistently exceeding expectations.

PROJECTS

[🔗 JukeBoxd](#) || *React, Tailwind CSS, TypeScript, Supabase, Spotify API*

- Designed and developed a social platform for music reviews, enabling users to share, rate, and discover music.
- Maintained active users with over **300 Accounts Created**, **2000+ Listens** and **150+ Ratings**.
- Integrated the **Spotify API** for seamless music discovery and **Supabase** for scalable database + authentication.
- Implemented responsive, mobile-first UI with **Tailwind CSS**, achieving **< 200ms** interaction latency in testing.
- Designed and implemented dynamic review feeds, and personalized recommendation algorithms.

[🔗 Ray Tracer](#) || *C++*

- Designed and built a high-performance C++ ray tracer, showcasing advanced object-oriented programming.
- Implemented efficient algorithms for ray-sphere and ray-plane intersections, with realistic reflection, refraction, and texture mapping features.
- Optimized the rendering pipeline with bounding volume hierarchies, achieving an **8x speedup** in render time.

[🔗 Fluid Dynamics](#) || *JavaScript, HTML/CSS*

- Developed a real-time Eulerian fluid simulation using JavaScript, modeling smoke advection and fluid flow.
- Implemented numerical solvers for pressure correction and advection to produce accurate, stable simulations.